

Kihwan Kim

Software Engineer · 7 years — building agentic AI into full-stack products

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Software engineer with 7 years of experience building products end to end — from React/TypeScript frontends into the server, and lately putting agentic AI into production. I built an agent that automates customer-support and QA operations by driving a browser across several internal tools, with a human-in-the-loop escape hatch for reliability. My background is in HCI and Visual Analytics; I care about making AI output something people can trust and act on. I work across the stack — Node.js (via Next.js), Redis for caching, and SQL databases (MySQL) — and pick up new backend technologies quickly (recently shipping Rust to an open-source game server).

Highlights

- **Agentic AI in production** — designed and shipped an agent that turns natural-language requests into browser actions across internal tools, cutting hours of manual customer-support investigation to minutes (Levvels, PageAgent)
- **Support operations & multi-tool integration** — connected account, payment, and mapping data scattered across several internal systems into one verifiable view, with a Slack human-in-the-loop hand-off so a stuck automation fails loudly to a person
- **Real-time, high-scale frontend** — designed a streaming state machine for LLM responses (send/complete/fail/regenerate) and optimized large-list rendering with windowing and dynamic heights on a large content platform (RIDI)
- **Full-stack ownership** — contributed 10 merged PRs to an open-source MMORPG, owning changes from the TypeScript client into the Rust server (cache eviction, account bans, trade validation)

Experience

Levvels — Software Engineer

Jul 2025 – Present

AI-native product development and internal operations automation, working across frontend and the agent/integration layer.

PageAgent-based Operations Automation agentic AI · CS/QA

- Automated repeated customer-support and QA investigation that required staff to carry state across customer chat, delivery tracking, internal admin, and email to make one decision
- Turned natural-language requests into reproducible browser action sequences and a single verifiable result view; operators can re-check each conclusion against its source
- Integrated account, payment, and mapping data from several internal tools into one comparison view — the same shape of work as CRM/tool integration
- Extended Alibaba's open-source PageAgent for a Chrome Managed Profile and added a Slack human-in-the-loop hand-off the base tool lacked, so failures surface to a person instead of returning a wrong result silently

AI Character Chat real-time streaming

- Separated in-progress streamed responses from persisted history; designed the send / complete / fail / regenerate state machine for interrupt-and-retry-heavy LLM responses over SSE
- Built authoring and review tooling for stories, lorebooks, and search keywords; reconciled specs across product docs, Figma, and Swagger

React · Next.js · Node.js · TypeScript · TanStack Query · Zustand · SSE · Playwright · Sentry · PageAgent

RIDI Corp. — Frontend Engineer

May 2022 – Jul 2025

Kept operational UI changes and experiments off the deployment critical path on a large, high-traffic content platform.

Server-driven UI & large-scale rendering

- Rendered server-declared UI on the client to decouple screen changes from deploys; migrated PHP pages toward React incrementally to lower risk
- Optimized large-list rendering with windowing, dynamic-height measurement, and scroll restoration; analyzed and reduced frame drops from large DOM trees
- Used Redis to cache server-driven UI payloads and reduce repeated backend load; tracked real-user errors with Sentry and stabilized mobile, desktop, and special-device environments

React · Next.js · TypeScript · Server-driven UI · Redis · Sentry · Virtualization

TmaxEnterprise — Research Engineer

Feb 2020 – May 2022

Made complex AI and analytics usable for non-experts — usability and explainability first.

AutoML Platform explainable AI · Python

- Built a codeless studio so users without model-development experience could build and operate models
- Developed explainable-AI visualizations that show the reasoning beside the prediction, so a non-expert has a basis to act

React · TypeScript · Material-UI · Python

Open Source

OpenMMO — Contributor

Jul 2026 – Present

- 10 PRs merged, owning changes end-to-end from the TypeScript client into the **Rust server** — cache eviction, account bans, and trade validation — while learning a new backend language on a live codebase
- Paired with a coding agent under a builder/reviewer workflow: the builder writes, a separate model reviews, and behavior is pinned by tests before submission

vite-plus — Contributor

Aug 2026

- Fixed a shell-integration gap in Rust (working-directory flag dropped before the env command) and covered the four shells' templates with snapshot tests (#2508, merged) — VoidZero, the team behind Vite and Vitest

Education

Ulsan National Institute of Science and Technology (UNIST)

2013 – 2020

M.S. in Computer Science (HCI · Visual Analytics; human-AI collaboration).

B.S. in Technology Management (CS multi-disciplinary major).

- “An Empirical Analysis on Transparent Algorithmic Exploration in Recommender Systems” — **CoRR 2021**
· arxiv.org/abs/2108.00151
- “Modeling Exploration/Exploitation Decisions through Mobile Sensing for Understanding Mechanisms of Addiction” — **MobiSys '19** · dl.acm.org/doi/10.1145/3307334.3328599

Full CV, case studies, and portfolio at kihwan.kim.